

# **Detection Engineering for Data Loss Prevention: Designing Splunk-Based Alerts and Response Plans for Data Exfiltration Incidents**

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## **Progress Report # 3**

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## **Problem Statement**

Data exfiltration incidents, whether driven by external attackers or malicious insiders, represent a critical threat to the confidentiality and integrity of organizational information assets. As enterprises increasingly adopt cloud services, remote work models, and heterogeneous infrastructures, data can leave corporate boundaries through a wide variety of channels, including web uploads, cloud storage platforms, email, collaboration tools, and covert network protocols. Traditional Data Loss Prevention (DLP) solutions typically focus on content inspection and policy enforcement within a single channel, which often leads to fragmented visibility and an overwhelming number of isolated alerts.

Security Information and Event Management (SIEM) platforms such as Splunk are designed to aggregate logs and events from multiple security and IT systems, offering a centralized view of activity across the environment. However, many organizations underutilize this aggregated data for DLP-specific detection engineering and lack well-defined response processes tailored to data exfiltration scenarios. The core problem addressed by this project is the absence of a structured, Splunk-based framework that uses aggregated telemetry to detect diverse data-exfiltration behaviors and couples these detections with clear, actionable incident response plans.

This project seeks to design, implement, and evaluate a detection-engineering solution for DLP scenarios that operates on aggregated data within Splunk. The solution will specify detection use cases for multiple exfiltration vectors, define the required data and system architecture, implement correlation and alerting logic, and document response playbooks that guide Security Operations Center (SOC) analysts from alert generation through to containment and recovery.

## **Related Work**

### **Literature References**

Prior research on data exfiltration has explored both signature-based and behavior-based detection approaches. Network-centric studies analyze flow statistics, traffic volumes, and protocol-specific features to identify anomalous outbound behavior indicative of data theft. Host-centric and application-centric works focus on access patterns, file operations, and process behaviors. More recent research has applied statistical models and machine learning to detect deviations from established baselines, particularly for low-and-slow exfiltration attempts that evade simple volume thresholds.

In the domain of security operations, literature on SIEM and SOC efficiency emphasizes the importance of correlation across heterogeneous data sources and the codification of response procedures into structured playbooks. These works highlight that effective detection is not only a matter of signal quality but also of how well alerts integrate into the operational workflows of analysts, including triage, enrichment, and remediation steps.

## Other Products and Projects

Commercial DLP solutions and Cloud Access Security Brokers (CASBs) provide extensive policy-based capabilities for monitoring and controlling data movement across endpoints, email, web gateways, and cloud services. They often include pre-built policies for sensitive data types and regulatory categories, as well as dashboards and incident views for compliance reporting. However, these products typically generate alerts within their own consoles and may lack deep, out-of-the-box correlation with other infrastructure and security logs, leading to siloed incident handling.

SIEM and Extended Detection and Response (XDR) platforms commonly offer predefined rules and content packs for detecting suspicious outbound traffic patterns, abnormal cloud activity, or signs of DNS tunneling. These content packs can accelerate deployment but usually require customization to reflect an organization's specific environment, threat model, and risk tolerance.

## Critical Comparison with the Proposed Design

The proposed project design differs from existing work and products in three main respects. First, it explicitly frames DLP for data exfiltration as a detection-engineering problem within Splunk, rather than as a feature of a standalone DLP solution. Second, this project focuses on a coherent set of use cases that span multiple channels and ties them together through aggregated analytics. Third, the project tightly couples detection logic with concrete incident response playbooks, bridging the gap between analytics and operations.

## Requirements

### Functional Requirements

The system shall:

#### 1. Data Ingestion and Normalization

- Ingest network, proxy, or firewall logs providing source, destination, and byte-count fields.
- Ingest DNS logs capturing query names, query types, responses, and client identifiers.
- Ingest SaaS and office-suite audit logs for email and file operations, including events such as message send, forwarding, and mailbox export.
- Ingest DLP or related security logs that report policy violations, rule matches, and actions taken.

#### 2. Detection Use Cases

- Define at least four data-exfiltration use cases covering high-volume outbound transfers, unsanctioned destinations, email-based exfiltration, and DNS-based exfiltration.

#### 3. Detection Logic Implementation

- Aggregate events over configurable time windows by user, host, or other entity.
- Compute features such as total bytes, event counts, destination rarity, and deviation from historical baselines.
- Generate alerts or notable events with contextual details.

#### **4. Response Playbooks**

- For each detection use case, define an incident response playbook with initial validation steps, enrichment actions, containment and remediation, and escalation paths.

#### **5. Evaluation and Reporting**

- Execute controlled tests using synthetic or lab-generated data.
- Evaluate the rate of false positives against benign traffic scenarios.
- Compile results, lessons learned, and recommendations into a structured report.

### **Non-Functional Requirements**

- Performance: Detection searches must complete within an acceptable time frame for operational use.
- Scalability: The design must remain viable as data volumes grow, using efficient aggregations.
- Reliability: Detection logic and playbooks must function consistently, with clear error-handling strategies.
- Explainability: Each detection rule must be transparent with documented logic, thresholds, and limitations.
- Maintainability: All rules, playbooks, and configurations must be modular and version-controlled.
- Compliance and Privacy: The project must respect applicable privacy constraints, relying on anonymized or synthetic data when necessary.
- Under 500MB/day ingestion (stay in license).
- 100% uptime (<5 min ingestion delay).

### **Implementation Progress This Period**

Since the submission of Progress Report #2, the project has advanced significantly. The two foundational detection use cases (high-volume outbound transfers and unsanctioned cloud storage access) implemented in the previous phase have been refined, and two new detection use cases have now been fully implemented: DNS-based data exfiltration detection and email-based exfiltration detection. In addition, incident response playbooks have been drafted for all four detection use cases, and expanded testing has been conducted using an augmented synthetic dataset.

## Expanded Synthetic Dataset

The synthetic dataset has been significantly expanded to support testing of the new DNS and email detection use cases. The updated dataset now includes approximately 25,000 log events across four source types.

| Source Type          | Log Count | Purpose                                        |
|----------------------|-----------|------------------------------------------------|
| Proxy / Network Logs | 10,000    | High-volume and unsanctioned domain detections |
| DNS Query Logs       | 8,000     | DNS-based exfiltration detection               |
| Email Audit Logs     | 5,000     | Email-based exfiltration detection             |
| DLP Policy Logs      | 2,000     | Policy violation correlation                   |

Each log type was designed to include a mix of benign activity and embedded exfiltration events. The DNS log dataset, for example, includes normal resolution traffic alongside simulated DNS tunneling sessions characterized by high query frequency and encoded subdomains of anomalous length.

## Detection Use Case #3 – DNS-Based Data Exfiltration

DNS tunneling is a covert exfiltration technique in which data is encoded into DNS query names (subdomains) and exfiltrated via recursive DNS resolution to an attacker-controlled authoritative server. This technique exploits the fact that DNS traffic is frequently permitted through network perimeters without deep inspection.

### Detection Approach

The detection strategy focuses on two complementary signals:

- Abnormally high DNS query volume from a single host within a short time window.
- Presence of query names with unusually long subdomain strings, which are characteristic of base64- or hex-encoded payload data.

Both signals are evaluated per source host over a one-hour rolling window. Hosts exhibiting both elevated query counts and anomalous subdomain lengths are flagged as high-priority exfiltration candidates.

### SPL Query – DNS Volume Anomaly

```
index=dns_logs
| eval subdomain_len=len(query_name)
| stats count as query_count avg(subdomain_len) as avg_len by src_ip
| where query_count > 500 AND avg_len > 40
| sort - query_count
```

## SPL Query – Long Subdomain Detection

```
index=dns_logs
| eval subdomain_len=len(query_name)
| where subdomain_len > 60
| stats count as suspicious_queries dc(query_name) as unique_queries by
src_ip
| where suspicious_queries > 50
| sort - suspicious_queries
```

## Detection Results

| src_ip    | query_count | avg_subdomain_len | Assessment            |
|-----------|-------------|-------------------|-----------------------|
| 10.0.0.14 | 1,842       | 73.2              | Confirmed Tunneling   |
| 10.0.0.31 | 987         | 68.7              | Confirmed Tunneling   |
| 10.0.0.7  | 412         | 31.4              | Borderline – Reviewed |
| 10.0.0.5  | 210         | 22.1              | Benign                |

The detection successfully flagged two hosts (10.0.0.14 and 10.0.0.31) that were embedded in the dataset as DNS tunneling sources. The third host (10.0.0.7) generated a borderline alert due to a legitimate automated monitoring tool; this threshold will be adjusted in subsequent tuning. No other benign hosts triggered the rule.

## Detection Use Case #4 – Email-Based Data Exfiltration

Email remains one of the most common channels for insider data exfiltration. Malicious or negligent insiders may use mass forwarding rules, bulk sending to external recipients, or mailbox export operations to remove sensitive data from the organization. This detection use case targets abnormal outbound email behavior as a proxy for exfiltration activity.

## Detection Approach

Two complementary SPL rules were implemented:

- Mass external forwarding: Identifies users who have sent or forwarded a high number of emails to external domains within a single day.
- Large attachment volume: Identifies users whose outbound email attachments exceed a cumulative size threshold, indicating potential bulk document exfiltration.

## SPL Query – Mass External Email Forwarding

```
index=email_logs action=send OR action=forward
| where match(recipient, "@(?!company\.com)")
| stats count as external_sends dc(recipient) as unique_recipients
    values(subject) as subjects by user
| where external_sends > 50 OR unique_recipients > 30
| sort - external_sends
```

## SPL Query – Large Outbound Attachment Volume

```
index=email_logs action=send attachment_size>0
| where match(recipient, "@(?!company\.com)")
| stats sum(attachment_size) as total_attachment_bytes
    count as email_count by user
| where total_attachment_bytes > 104857600
| eval total_attachment_mb=round(total_attachment_bytes/1048576,1)
| sort - total_attachment_bytes
```

## Detection Results

| User              | External Sends | Unique Recipients | Attachment Vol. | Assessment              |
|-------------------|----------------|-------------------|-----------------|-------------------------|
| bob@company.com   | 147            | 89                | N/A             | Confirmed – Mass Fwd    |
| carol@company.com | 23             | 18                | 312 MB          | Confirmed – Bulk Attach |
| dave@company.com  | 61             | 12                | N/A             | Confirmed – Mass Fwd    |
| eve@company.com   | 8              | 6                 | 14 MB           | Benign                  |

Three exfiltration patterns were correctly identified from the synthetic dataset. The forwarding-based rule caught two simulated insider scenarios, while the attachment-volume rule flagged a simulated bulk document export. One benign user generated no alerts.

## Incident Response Playbooks

As planned, incident response playbooks have been drafted for all four detection use cases. Each playbook follows a consistent structure encompassing initial validation, enrichment, containment, and escalation. The four playbooks are summarized below.

### Playbook 1 – High-Volume Outbound Transfer

#### Step 1: Initial Validation

- Confirm the alert by reviewing the raw events behind the flagged src\_ip and domain pair.

- Determine whether the transfer volume falls within known scheduled backup or software update windows.
- Check asset inventory: is the source host a server with a legitimate bulk-transfer function?

### **Step 2: Enrichment**

- Query user-to-host mapping to identify the account associated with the source host.
- Review prior alerts for the same user or host in the past 30 days.
- Classify the destination domain (sanctioned, unsanctioned, or unknown).

### **Step 3: Containment**

- If exfiltration is confirmed: block outbound access to the destination domain at the web proxy or firewall.
- Optionally isolate the host from the network pending further investigation.
- Notify the data owner and CISO as appropriate.

### **Step 4: Escalation and Documentation**

- Open a formal incident ticket; assign severity based on data sensitivity and volume.
- Preserve relevant logs and alert artifacts for forensic analysis.
- Notify legal and compliance teams if regulated data is involved.

## **Playbook 2 – Unsanctioned Cloud Storage Access**

### **Step 1: Initial Validation**

- Verify the alert by confirming the destination domain appears in the unsanctioned domain lookup.
- Determine if the upload was performed via browser, sync client, or API.
- Assess whether the activity is covered by an approved business exception.

### **Step 2: Enrichment**

- Identify the user associated with the source IP.
- Review the user's job function and whether cloud storage access is necessary for their role.
- Correlate with DLP policy logs for any simultaneous content-inspection alerts.

### **Step 3: Containment**

- Block access to the unsanctioned domain via web proxy policy.
- Initiate a user interview through the HR/security team.
- Request account access review if suspicious intent is suspected.

#### **Step 4: Escalation and Documentation**

- Escalate to security management if sensitive data types (PII, financial, IP) are indicated.
- Document all findings, actions, and outcomes in the incident tracking system.

### **Playbook 3 – DNS-Based Exfiltration**

#### **Step 1: Initial Validation**

- Review the raw DNS query logs for the flagged host to manually inspect subdomain patterns.
- Determine whether the queried domains resolve to known benign infrastructure (e.g., CDN, monitoring tools).
- Calculate entropy of subdomain strings as additional confirmation of encoded payload traffic.

#### **Step 2: Enrichment**

- Identify the user logged onto the flagged host at the time of the detected activity.
- Cross-reference the contacted domain with threat intelligence sources if available.
- Review network flow logs to confirm outbound traffic correlated with DNS queries.

#### **Step 3: Containment**

- Block the destination domain at the DNS resolver or firewall.
- Isolate the host from the network to prevent continued exfiltration.
- Capture a memory image and disk image of the host for forensic investigation.

#### **Step 4: Escalation and Documentation**

- DNS tunneling activity typically indicates a sophisticated threat; escalate immediately to the incident response team.
- Preserve all DNS logs and network captures as evidence.
- Engage legal counsel if external attackers or criminal activity is suspected.

### **Playbook 4 – Email-Based Exfiltration**

#### **Step 1: Initial Validation**

- Review the email audit logs for the flagged user to confirm the forwarding or sending pattern.
- Assess whether an auto-forwarding rule exists in the user's mailbox.

- Classify recipient domains to determine if all destinations are expected business contacts.

### **Step 2: Enrichment**

- Pull the user's prior email activity to establish a behavioral baseline.
- Review recent HR and access activity for anomalies (e.g., resignation notice, performance issues).
- Correlate with endpoint DLP alerts that may indicate prior file staging activity.

### **Step 3: Containment**

- Disable the auto-forwarding rule if one exists.
- Temporarily block outbound email to non-company domains pending investigation.
- Request legal hold on the user's mailbox.

### **Step 4: Escalation and Documentation**

- Notify HR, legal, and CISO as appropriate.
- Preserve email logs and attachment copies as evidence.
- Follow organizational procedures for potential insider threat cases.

## **Test Plan**

### **Test Objectives**

- Validate that each detection rule correctly identifies its targeted exfiltration scenario.
- Assess the rate and nature of false positives under benign conditions.
- Confirm that response playbooks provide clear and actionable guidance to analysts.

### **Test Types**

#### **1. Unit-Level Detection Tests**

- Construct small, controlled datasets for each use case where the expected number and nature of alerts are known.
- Execute detection searches and compare outputs against expectations.

#### **2. Scenario-Based Tests**

- Simulate end-to-end exfiltration scenarios and verify the complete chain: data ingestion → detection → alert generation → engagement of playbook steps.

#### **3. Baseline and False-Positive Tests**

- Apply detection rules to benign datasets representing normal user activity.

- Record and analyze alerts generated, adjusting thresholds and filters where necessary.

#### **4. Performance Tests**

- Measure query execution times for typical look-back periods and data volumes.

## **Risk Plan**

### **Identified Risks and Mitigations**

#### **R1: Limited Realistic Data**

- Risk: Lack of diverse, real-world logs may reduce the realism of detections and evaluations.
- Mitigation: Use carefully designed synthetic datasets informed by known exfiltration behaviors; clearly document limitations and assumptions. Status: Ongoing – dataset expanded in this phase.

#### **R2: High False-Positive Rate**

- Risk: Overly simplistic detection rules may generate excessive alerts.
- Mitigation: Employ baselining, per-entity thresholds, contextual filtering, and iterative tuning. Status: One borderline DNS alert identified; threshold adjustment planned.

#### **R3: Time Constraints**

- Risk: Insufficient time to implement all planned use cases or advanced analytics.
- Mitigation: Core four use cases are now complete; remaining work focuses on tuning and documentation. Status: On schedule.

#### **R4: Tooling or Integration Limitations**

- Risk: Inability to integrate certain data sources or response systems in the lab environment.
- Mitigation: Conceptual playbook designs documented; implementation focused on available components. Status: Managed.

#### **R5: Complexity of Detection Logic**

- Risk: Overly complex rules may be difficult to understand and maintain.
- Mitigation: Detection logic kept interpretable; all SPL queries documented with inline comments. Status: Managed.

# Implementation Status

## Phased Implementation Overview

| Phase   | Description                                      | Weeks | Status      |
|---------|--------------------------------------------------|-------|-------------|
| Phase 1 | Environment Setup and Requirements Realization   | 1–3   | Complete    |
| Phase 2 | Data Model and Feature Design                    | 3–4   | Complete    |
| Phase 3 | Detection Implementation (UC1 & UC2)             | 5–7   | Complete    |
| Phase 4 | Detection Implementation (UC3 & UC4) + Playbooks | 8–9   | Complete    |
| Phase 5 | Testing, Tuning, and Evaluation                  | 10–12 | In Progress |
| Phase 6 | Documentation and Finalization                   | 13–14 | Upcoming    |

## Completion Status – Current Period

- Implemented DNS-based exfiltration detection (UC3) – two complementary SPL rules operational.
- Implemented email-based exfiltration detection (UC4) – mass forwarding and bulk attachment rules operational.
- Expanded synthetic dataset to ~25,000 events across four source types.
- Drafted incident response playbooks for all four detection use cases.
- Conducted initial scenario-based and unit-level testing for UC3 and UC4.
- Identified one borderline false-positive in DNS detection; threshold adjustment in progress.

## Project Timeline

### Task Breakdown

Representative tasks and current status:

| Task | Description                                           | Status   |
|------|-------------------------------------------------------|----------|
| T1   | Configure data inputs, indexes, and field extractions | Complete |
| T2   | Define use-case templates and threat models           | Complete |

| Task | Description                                                | Status                        |
|------|------------------------------------------------------------|-------------------------------|
| T3   | Implement feature extraction searches and baselines        | Complete                      |
| T4   | Implement correlation searches for each detection use case | Complete                      |
| T5   | Draft and refine incident response playbooks               | Complete – Refinement ongoing |
| T6   | Develop synthetic data and test scenarios                  | Complete – Expansion ongoing  |
| T7   | Execute tests and collect metrics                          | In Progress                   |
| T8   | Compile documentation, diagrams, and final report sections | In Progress                   |

## Milestone Timeline

| Weeks | Milestone                                                           | Status      |
|-------|---------------------------------------------------------------------|-------------|
| 1–2   | Requirements finalization and initial Splunk setup                  | Complete    |
| 3     | Data ingestion for first core log type; conceptual data model draft | Complete    |
| 4–5   | Feature extraction searches; UC1 and UC2 detection rules            | Complete    |
| 6–7   | UC3 and UC4 detection rules; initial playbook drafts                | Complete    |
| 8–9   | Playbook finalization; expanded synthetic datasets                  | Complete    |
| 10–11 | Full test plan execution; threshold tuning; results documentation   | In Progress |
| 12–14 | Final report preparation; presentation materials                    | Upcoming    |

## Preliminary Results Summary

The following table summarizes detection performance across all four implemented use cases based on testing conducted to date.

| Detection Rule                   | Alerts Generated | Expected Alerts | False Positives | Status          |
|----------------------------------|------------------|-----------------|-----------------|-----------------|
| UC1 – High-Volume Transfer       | 4                | 4               | 0               | Validated       |
| UC2 – Unsanctioned Domain Access | 6                | 6               | 0               | Validated       |
| UC3 – DNS-Based Exfiltration     | 3                | 2               | 1 (borderline)  | Tuning Required |
| UC4 – Email-Based Exfiltration   | 3                | 3               | 0               | Validated       |

Overall detection accuracy across validated use cases remains high. The single borderline alert in UC3 has been attributed to a threshold sensitivity issue with automated monitoring tools and will be addressed in the next tuning cycle. No other false positives were observed across any of the benign dataset samples tested during this phase.

## Challenges and Observations

Several challenges were encountered and observations made during this reporting period:

- DNS subdomain length alone is an imperfect signal; combining it with query volume substantially reduces the false-positive rate. Future tuning will incorporate entropy scoring of subdomain strings as a third signal.
- Email detection thresholds required careful calibration. An initial threshold of 20 external sends per day generated several false positives from users in sales and marketing roles. Raising the threshold to 50 sends and requiring concurrent involvement of more than 30 unique recipients substantially improved precision.
- Playbook design highlighted the need for pre-populated lookup tables of sanctioned domains and authorized cloud services, as manual classification during triage is time-consuming.
- Expanding the synthetic dataset confirmed that detection logic scales well to higher event volumes without significant query performance degradation.

## Next Steps

The final phase of the project will focus on completing the test plan, refining all detection rules, and consolidating the full project documentation.

Planned tasks for the next period include:

- Complete the full scenario-based test plan execution, including benign false-positive testing across all four detection use cases.

- Tune the DNS detection threshold using entropy-based scoring to reduce the borderline false-positive rate.
- Finalize and version-control all SPL queries, lookup tables, and playbook documentation.
- Compile detection performance metrics, threshold-tuning logs, and qualitative observations into the evaluation chapter of the final report.
- Prepare architecture diagrams and flow diagrams for playbooks to include in the final report.
- Begin drafting the final project report and presentation slides.

## Conclusion

Progress Report #3 documents the successful implementation of the two remaining detection use cases (DNS-based and email-based exfiltration) and the completion of incident response playbooks for all four use cases defined in the project scope. All four detection rules have been tested against the expanded synthetic dataset and demonstrate high accuracy, with one threshold adjustment pending for the DNS detection use case.

The project is on schedule. The next and final phase will focus on comprehensive test execution, detection tuning, and preparation of the final report and presentation. The detection-engineering framework developed in this project is positioned to produce a complete, operationally relevant set of Splunk-based detection artefacts and response playbooks for data exfiltration scenarios.

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